# VITyarthi – Build Your Own Project

# Smart Grocery Store Inventory & Billing System

CSE1021 – Problem Solving and Programming

Python Console Application

# 1. Introduction

Smart Grocery Store Inventory & Billing System is a console-based Python application for managing grocery products, shopping carts, billing and inventory analysis.

# 2. Problem Statement

Small grocery stores need a simple method to maintain product information, calculate bills and monitor stock. This project provides a modular Python solution for these operations.

# 3. Objectives

Manage products, search products, maintain a shopping cart, generate bills, apply discounts, update inventory and produce inventory reports.

# 4. Functional Requirements

Inventory management; product search; shopping cart management; bill generation; discount calculation; inventory update; inventory analysis; sorting report.

# 5. Non-Functional Requirements

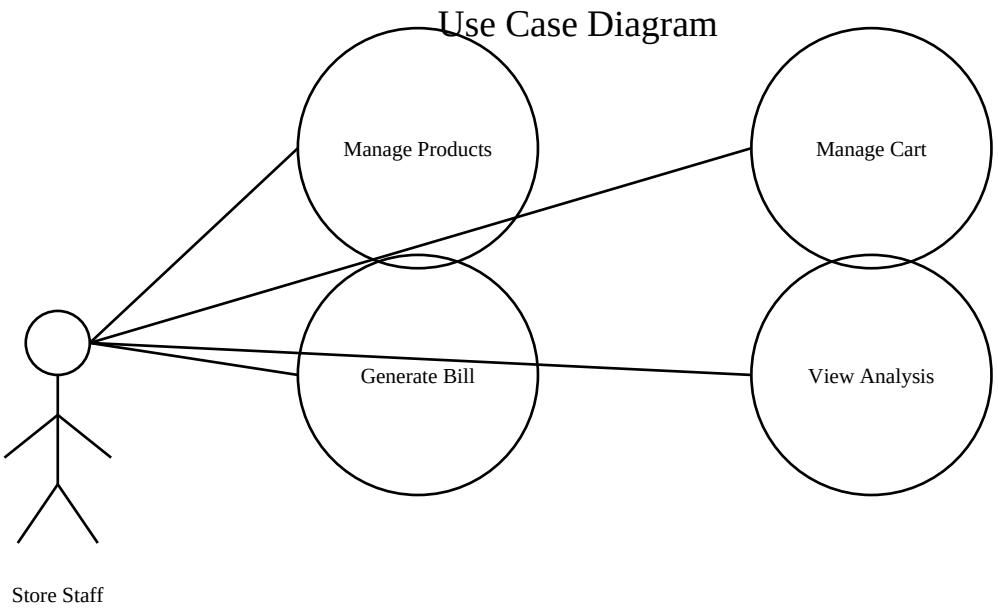
Usability through a menu-driven interface; reliability through input validation; maintainability through separate modules; resource efficiency through built-in Python data structures; error handling for invalid inputs.

# 6. System Architecture

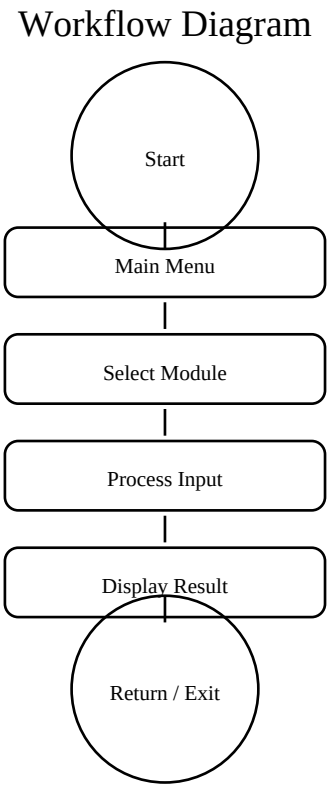
The application uses a modular console architecture. main.py controls the workflow and calls inventory.py, cart.py, billing.py and analysis.py. validation.py handles input validation and data.py contains initial product data.

# 7. Design Diagrams

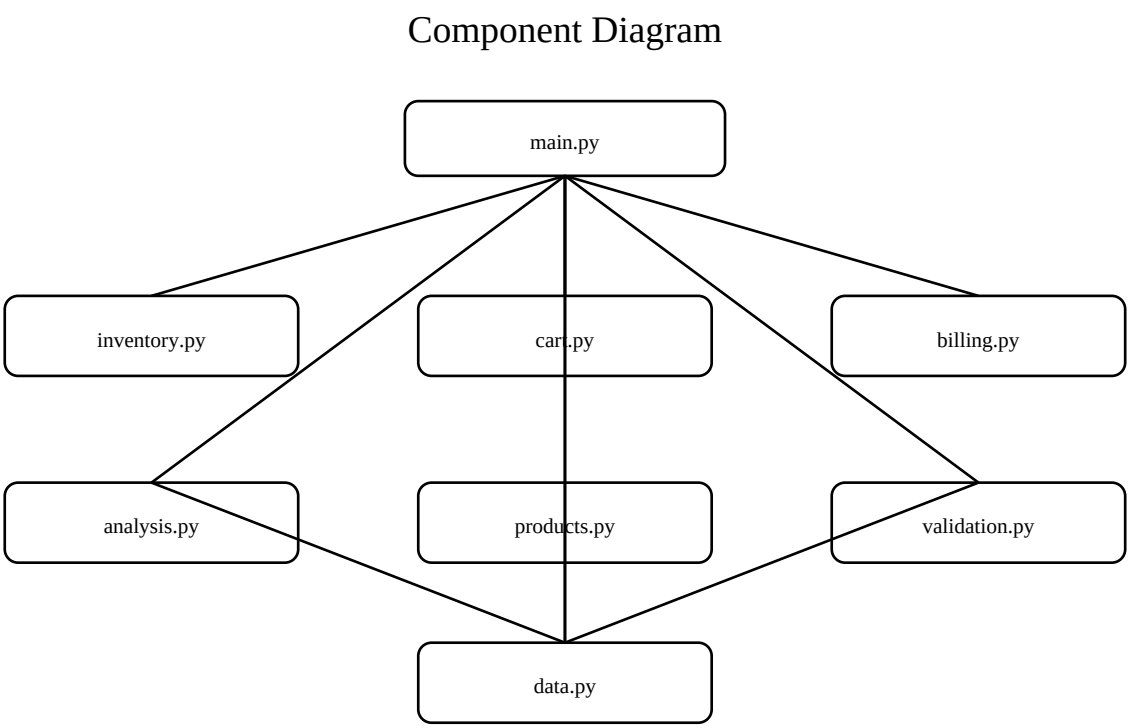
## 7.1 Use Case Diagram



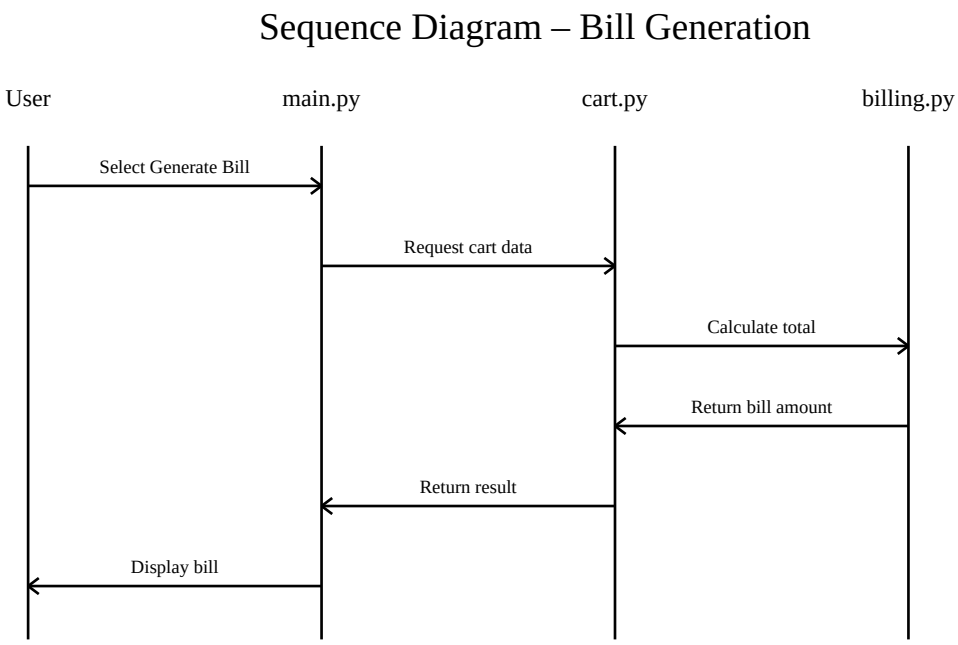

## 7.2 Workflow Diagram



## 7.3 Component Diagram




## 7.4 Sequence Diagram



## 8. Data / Storage Design

The project uses in-memory Python dictionaries and lists rather than an external database. Each product ID maps to a dictionary containing name, price and stock. Because there is no external database, an ER diagram is not applicable.

## 9. Design Decisions & Rationale

A console application keeps the implementation focused on the Python programming concepts covered in the course. Dictionaries provide key-value lookup, lists support collections and processing, loops support repeated operations, and functions provide modularity.

## 10. Implementation Details

Inventory operations add, update, remove and search products. The cart stores product IDs and quantities. Billing calculates subtotal, discount and final amount. Analysis calculates stock and inventory statistics.

## 11. Screenshots / Results

The repository contains sample_output.txt. Before submission, run the application and capture screenshots of the main menu, inventory table, shopping cart, final bill and inventory analysis, then place them in the report if required by the portal.

## 12. Testing Approach

The tests cover subtotal calculation, discount calculation, total stock, inventory value and low-stock identification. Run python tests/test_project.py.

## 13. Challenges Faced

Stock consistency between the shopping cart and inventory was handled by checking availability before adding items and updating stock only after bill generation.

## 14. Learnings & Key Takeaways

The project demonstrates modular Python programming, lists, dictionaries, loops, conditional statements, functions, searching, sorting, input validation and arithmetic processing.

## 15. Future Enhancements

Possible future additions include persistent file storage, customer accounts, barcode support, database integration and a graphical interface.

## 16. References

VITyarthi – Build Your Own Project: General Project Instructions & Submission Guidelines. CSE1021 Problem Solving and Programming course material, Units 2–5.